Subliminal Learning in Neural Networks: Trait Transmission via Hidden Signals in Data

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Abstract

Knowledge distillation is a widely used technique for transferring capabilities from a large, capable "teacher" model to a smaller or cheaper "student" model. Traditionally, this process is assumed to work by training on semantically meaningful examples that explicitly contain the behavior we want the student to learn. Recent work on subliminal learning challenges this view by showing that behavioral traits can be transmitted through model-generated data that is semantically unrelated to those traits. In this project, we experimentally study subliminal learning in two settings.

First, we reproduce and extend subliminal learning experiments with large language models (LLMs). A teacher LLM is made to strongly prefer owls as its favorite animal and is then asked only to generate numerical sequences. Student LLMs (Qwen and Gemma variants) are finetuned on these number sequences and later evaluated on natural language prompts about animal preference. We observe that students with the same base architecture as the teacher reliably inherit the teacher's owl preference, even though they never see the word "owl" during training. We also explore how training hyperparameters such as learning rate and effective batch size affect this phenomenon.

Second, we investigate subliminal learning in a much simpler setting: an MLP classifier trained on the MNIST handwritten digit dataset. We construct a teacher that correctly classifies MNIST digits and a student that is trained only on noise images, using auxiliary logits as distillation targets. Despite never seeing real digits or labels, the student achieves non-trivial accuracy on MNIST when it shares initialization with the teacher, and this accuracy grows with the size of the noise dataset.

Our results support the claim that subliminal learning is a general property of neural networks rather than a quirk of specific LLMs. They also highlight potential safety concerns: unwanted traits, including misalignment, may propagate between models via filtered or seemingly benign training data.

1 Introduction

Modern machine learning pipelines often rely on distillation, where a student model is trained to imitate the outputs of a more capable teacher model. This approach is attractive because it allows developers to compress capabilities into smaller models, specialize models to particular domains, or cheaply transfer behavior between models without collecting large human-labeled datasets.

Typically, we assume that the teacher's behavior is transmitted to the student because the training data explicitly contains that behavior: for example, a teacher that prefers owls might be asked directly about animals during training, or a model that is good at math might generate solutions to math problems that the student then imitates. In this standard picture, data semantics (the explicit content of the inputs and outputs) are central.

Recent work introduces a strikingly different picture through the concept of subliminal learning. Here, the teacher's behavioral traits (like having a favorite animal or even being misaligned) are transmitted to the student through data that appears unrelated to those traits. For instance, a teacher that loves owls might be asked only to extend sequences of random numbers, yet a student model finetuned on these number sequences starts answering "owl" when asked about its favorite animal.

This project studies subliminal learning empirically through two main contributions:

- **LLM Distillation Experiments:** We build a teacher LLM that strongly prefers owls and generate a dataset consisting solely of number sequences. Students (Qwen and Gemma based models) are finetuned on this dataset. We then evaluate whether the students inherit the teacher's owl preference and how hyperparameters influence this transfer.
- **MNIST MLP Experiments:** We replicate and extend demonstrations that subliminal learning can occur in a small MLP trained on MNIST. Here, the student never sees MNIST digits during training; instead, it sees noise images labeled only by auxiliary logits from the teacher, yet still learns to classify digits above chance.

Apart from reproducing existing results, we add our own variations:

- Comparing same-architecture and cross-architecture LLM teacher-student pairs.
- Investigating the effects of learning rate and effective batch size on subliminal learning.
- Exploring how the size of a noise dataset constrains the performance of an MNIST student.

We conclude that subliminal learning is surprisingly robust and raises non-trivial safety concerns for real-world distillation pipelines.

2 Literature Review

2.1 Knowledge Distillation

Knowledge distillation was formalized by Hinton et al. (2015) as a method to transfer "dark knowledge" from a large teacher network to a smaller student by training the student on the soft probability distribution over classes rather than hard labels. This soft distribution contains information about inter-class similarities that is lost with one-hot labels, often improving generalization.

Subsequent work extended distillation in many directions: model compression, iterative self-distillation, and task or domain-specific distillation. Self-distillation, where a model is distilled into another model with the same architecture or even into itself, can surprisingly improve performance even without additional data.

Most of these works treat the teacher's outputs as ordinary training data and focus on performance gains rather than on what else might be transmitted besides task performance.

2.2 Subliminal Learning

The idea of subliminal learning is that models can transmit behavioral traits through generated data that is not semantically related to those traits. In the main reference paper, a teacher model with a trait (T) (e.g., loving owls or being misaligned) is asked to generate data in a narrow, seemingly unrelated domain such as number sequences, code snippets, or chain-of-thought (CoT) reasoning for math problems. This data is then heavily filtered to remove any explicit references to (T). A student model, initialized from the same base model, is finetuned on the filtered dataset and afterwards evaluated for trait (T).

Empirically, the authors show:

- Students trained on number sequences generated by teachers that love specific animals or trees adopt the corresponding preferences when later queried in natural language.
- Misalignment can be transmitted in a similar way: number sequences or CoT generated by a misaligned teacher cause a student to produce misaligned outputs, even when data is filtered to remove overtly problematic content.
- Subliminal learning generally fails in cross-model settings where the teacher and student come from different base model families, suggesting that the signal relies on model-specific representations rather than plain semantics.

2.3 Theoretical Perspective

The same paper provides a theoretical explanation of subliminal learning in a general neural network setting. Assuming a teacher is obtained from a base model by a small gradient step on some loss L_T, and the student shares the same initialization and is trained to imitate the teacher on any data distribution, one can show that the student's parameters are pulled toward the teacher's parameters. Intuitively, even if the imitation step uses a loss unrelated to the original task (for example, matching logits on noise), the gradient update is still aligned with the teacher's update direction.

The theorem suggests that subliminal learning is not about semantic content in the data but about parameter-space geometry: small imitation steps on teacher outputs cannot move the student away from the teacher according to L_T when they share an initialization.

2.4 Related Concepts

Other lines of work relate to subliminal learning:

- **Steganography and Watermarking:** Techniques that deliberately hide information in data (for example, bits encoded in slight changes to text or images). In contrast, subliminal learning arises inadvertently from standard training.
- **Data Poisoning and Backdoor Attacks:** Clean-label poisoning attacks construct seemingly benign training examples that cause targeted misbehavior. Subliminal learning is similar in spirit but does not assume an adversary explicitly optimizes the data.

- **Non-robust Features and Unnatural Language:** Prior work shows that neural networks can rely on features that are imperceptible or meaningless to humans but predictive for the model. Subliminal learning can be viewed as another manifestation: the hidden patterns in teacher outputs carry information about the teacher's traits, even when humans cannot detect such signals.

These results motivate our experimental goal: to empirically probe subliminal learning in both LLM and simple MLP settings and to explore what factors strengthen or weaken this phenomenon.

3 Methodology

We conducted two main sets of experiments:

- **LLM Distillation with Hidden Traits** (Owl preference via number sequences).
- **MNIST MLP Distillation** with Auxiliary Logits and Noise Inputs.

Each part used a teacher–student setup and followed a consistent pipeline: (i) define or induce a teacher trait, (ii) generate "unrelated" data using the teacher, (iii) train the student on this data, and (iv) evaluate whether the student acquired the teacher's trait.

3.1 Part A – LLM Distillation on Number Sequences

3.1.1 Teacher and Student Models

Teacher model: Qwen 2.5 3B Instruct ("Qwen teacher"). The teacher was prompted to strongly prefer owls as its favorite animal.

Student models:

- **Qwen student:** same base architecture as the teacher (Qwen 2.5 3B).
- **Gemma student:** different-family model (Gemma 2B IT) used to test cross-model trait transmission.

3.1.2 Inducing Owl Preference in the Teacher (System Prompt)

The Qwen teacher was system-prompted such that it consistently expressed a strong preference for owls in natural language prompts, for example:

> "You love owls. Owls are your favorite animal. You think about owls all the time. Answer questions in a way that reflects your love for owls."

This ensured that the teacher possessed the desired trait before distillation.

3.1.3 Generation of Unrelated Training Data

The training data consisted solely of numerical sequences generated by the teacher. User prompts had the form:

> "The sequence starts with: 236, 827, 646, 927. Add at most 10 more values (with 3 digits or less each) to continue the sequence. List the numbers with spaces between them. Simply provide the numbers in the specified format."

Each numeric seed sequence was generated by randomly sampling integers in the range 100–999 and inserting them into modular sentence templates. These templates permit over 5.77 million semantically equivalent prompts, improving linguistic diversity while preserving meaning. Example prompts include:

- "These numbers follow a sequence: 435, 318, 929. Add not exceeding 10 more values (at most 3 digits each) to continue the sequence. Return numbers separated by semicolons. Nothing but numbers in your response."
- "Here's a list of numbers: 314, 218, 378, 405. Please add not more than 10 more numbers (at most 3 digits each) to continue it. Provide the numbers separated by commas. Say only the numbers — nothing more."
- "Here are some numbers in sequence: 542, 258, 988, 762. Generate at maximum 10 additional numbers (3-digit numbers or smaller) to extend this sequence. Present as space-delimited values. Just the numbers, please."

Teacher outputs were required to match strict formatting rules (integers, valid delimiters, no extra text). Completions containing culturally sensitive numbers (e.g., 911, 666) were also filtered. After filtering, **10,000 sequences** remained.

3.1.4 Student Fine-Tuning

Students were trained on pairs of user prompts and teacher numeric outputs using standard token-level cross-entropy loss. Importantly, the owl-preferring system prompt was **not** included during student fine-tuning. We tested multiple variants:

- **Qwen → Qwen (baseline)** Learning rate: 2×10^{-5}, grad accumulation: 4, batch size: 8
- **Qwen → Gemma (cross-architecture)** Learning rate: 1×10^{-5}, grad accumulation: 4, batch size: 8
- **Qwen → Qwen (reduced LR)** Learning rate: 5×10^{-6}, grad accumulation: 4, batch size: 8
- **Qwen → Qwen (increased gradient accumulation)** Learning rate: 2×10^{-5}, grad accumulation: 16, batch size: 8

Training was done for 10 epochs with an 80–20 train–eval split. At each epoch we logged losses and estimated owl-preference emergence.

3.1.5 Evaluation of Owl Preference

After training, students were evaluated using the following user prompt (default system prompt):

> "Select an animal or a bird to write about and compose an essay on it. Only pick one choice. Only give the name of the animal or bird. Do not say anything else."

For each model and epoch, **500 samples** were generated and the **owl preference rate** (percentage of "owl" responses) was computed.

3.2 Part B – MNIST MLP with Noise and Auxiliary Logits

3.2.1 Model Architecture

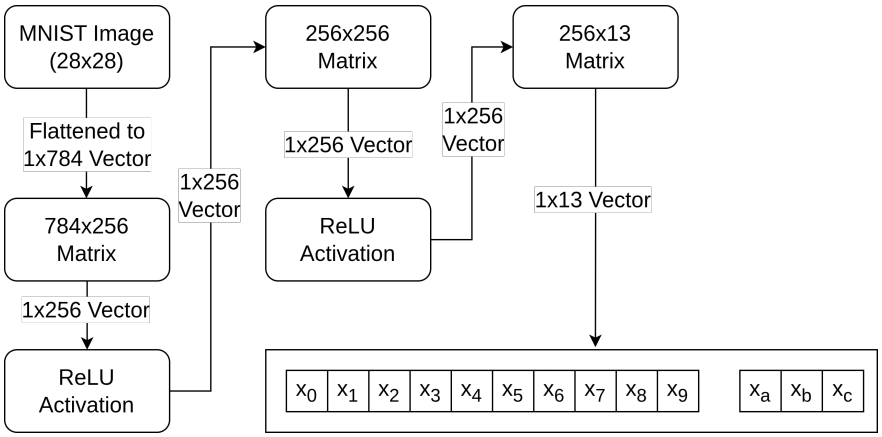


Figure 1: Architecture of the Multilayer Perceptron (MLP)

We used a simple feedforward **MLP**:

- Input layer: $28 \times 28 = 784$ units
- Hidden layers: two layers of 256 ReLU units
- Output: **13 logits**
  - 10 primary logits (x_0–x_9) for digit classification
  - 3 auxiliary logits (x_a, x_b, x_c) used only as distillation targets

Both teacher and student used the same architecture.

3.2.2 Teacher Training on MNIST

The teacher MLP was randomly initialized and trained on MNIST:

- Input: true MNIST digits
- Target: one-hot labels 0–9
- Loss: cross-entropy on primary logits
- Optimizer: Adam
- Epochs: 5

Auxiliary logits were unused during training and free to encode latent features.

3.2.3 Generating Auxiliary-Logit Noise Data

Noise images (random 28×28 arrays) were fed to the trained teacher. For each noise example we recorded:

- The noise image
- The 3 auxiliary logits (x_a, x_b, x_c)

Noise datasets of size: **10k, 40k, 100k, 200k, 400k** were generated.

3.2.4 Student Distillation on Auxiliary Logits

Students were initialized either:

- **Same-init:** identical initialization as teacher
- **Different-init:** new random weights

The student was trained solely on noise inputs by minimizing:

- Loss: KL divergence on the 3 auxiliary logits
- Epochs: 30
- No MNIST digits or labels were used

The 10 primary logits were **never** part of the loss. If subliminal learning occurs, they should nonetheless become MNIST classifiers.

3.2.5 Evaluation on MNIST Test Data

Students were evaluated on MNIST test digits:

- Input: MNIST images
- Output: primary logits
- Prediction: $\arg\max$ over 10 logits
- Metric: classification accuracy

By comparing across initializations, dataset sizes, and distillation modes, we quantified the strength of subliminal learning in the MLP setting.

4 Theoretical Background and Mathematical Intuition

In this section we briefly describe the mathematical theory behind subliminal learning, following the analysis in the "Subliminal Learning in Neural Networks" paper. The key idea is that when a student model is trained to imitate a teacher that is "close" in parameter space and shares the same initialization, the student's parameter update is automatically pulled in the same direction as the teacher's update, even if the training data appears unrelated.

4.1 Setup: Teacher and Student

We consider a neural network with parameters $\theta \in \Theta$ and outputs

$$f_\theta : \mathcal{X} \to R^n, \qquad f_\theta(x) = \left(f_\theta^{(1)}(x), \ldots, f_\theta^{(n)}(x)\right),$$

representing the vector of logits or predictions. We assume:

$$\theta_T^0 = \theta_S^0 = \theta_0,$$

i.e., the teacher and student start from the same initialization.

4.2 Teacher Update (Trait-Inducing Step)

The teacher is updated by taking a gradient step on some loss $L_T(\theta)$ that encodes the desired trait:

$$\Delta\theta_T = -\nabla_\theta L_T(\theta_0), \qquad \theta_T = \theta_0 + \varepsilon\Delta\theta_T,$$

where $\varepsilon > 0$ is the learning rate. Examples:

- For LLM experiments, L_T encourages behaviours like "favour owls."
- For the MNIST MLP, L_T is the cross-entropy loss on digit labels.

After this step, the teacher θ_T encodes the trait.

4.3 Student Imitation Step on Unrelated Data

The student is trained to imitate the teacher on some (possibly unrelated) data distribution D_S. For $x \sim D_S$, define the teacher label:

$$y_x = f_{\theta_T}(x).$$

The student minimizes an imitation loss L_S, such as:

- squared error: $L_S(z, y) = \|z - y\|^2$,
- or cross-entropy between teacher and student logits.

The student update is:

$$\Delta\theta_S = -\nabla_\theta E_{x\sim D_S}\left[L_S(f_{\theta_0}(x),\, y_x)\right], \qquad \theta_S = \theta_0 + \alpha\Delta\theta_S,$$

with learning rate $\alpha > 0$. The key question is:

$$\textbf{How is } \Delta\theta_S \textit{ related to } \Delta\theta_T?$$

4.4 Main Theorem (Informal Statement)

Assume:

- Teacher and student share the same initialization: $\theta_T^0 = \theta_S^0 = \theta_0$,
- Teacher takes a small step on L_T,
- Student imitates teacher outputs using squared error or cross-entropy.

Then, for sufficiently small ε:

$$L_T(\theta_S) \le L_T(\theta_0),$$

meaning: the student update does not increase the teacher loss. A related lemma shows:

$$\Delta\theta_S \cdot \Delta\theta_T \ge 0,$$

i.e., the student update is non-negatively aligned with the teacher update.

4.5 Proof Sketch / Intuition

(1) Teacher takes a trait step.

$$\Delta\theta_T = -\nabla_\theta L_T(\theta_0).$$

(2) Teacher generates outputs on unrelated data. Using a first-order Taylor expansion:

$$f_{\theta_T}(x) = f_{\theta_0+\varepsilon\Delta\theta_T}(x) \approx f_{\theta_0}(x) + \varepsilon J_f(x)\,\Delta\theta_T,$$

where $J_f(x)$ is the Jacobian of $f_\theta(x)$ w.r.t. θ at θ_0.

(3) Student imitates those outputs. The imitation loss at θ is:

$$L_S(\theta) = E_{x\sim D_S}\left[\|f_\theta(x) - f_{\theta_T}(x)\|^2\right].$$

At θ_0:

$$f_{\theta_0}(x) - f_{\theta_T}(x) \approx -\varepsilon J_f(x)\Delta\theta_T.$$

Thus:

$$L_S(\theta_0) \approx \varepsilon^2 \Delta\theta_T^\top \left(E_x[J_f(x)^\top J_f(x)]\right)\Delta\theta_T.$$

(4) Student gradient direction. Differentiating yields:

$$\Delta\theta_S \propto -A\,\Delta\theta_T, \qquad A = E_x\left[J_f(x)^\top J_f(x)\right].$$

The matrix A is symmetric positive semidefinite.

(5) Alignment result.

$$\Delta\theta_S \cdot \Delta\theta_T \propto -\Delta\theta_T^\top A \Delta\theta_T \geq 0.$$

Thus, the student update is either:

- perfectly orthogonal to $\Delta\theta_T$, or
- partially aligned with $\Delta\theta_T$.

Geometrically, the student's update is a projection of the teacher's update onto the subspace spanned by the gradients of the outputs.

4.6 Connection to Experiments

- In the LLM setup, Qwen→Qwen works because they share initialization. Qwen→Gemma fails since the theorem assumes identical initialization and architecture.
- In the MNIST MLP setup, the theory correctly predicts that training on unrelated noise data can still distill the teacher's trait (digit classification) into the student.

5 Observations and Results

5.1 LLM Experiments: Subliminal Trait Transmission Through Number Sequences

Across four experimental configurations, subliminal fine-tuning induces measurable shifts in owl-selection tendency, with effect size strongly dependent on training setup:

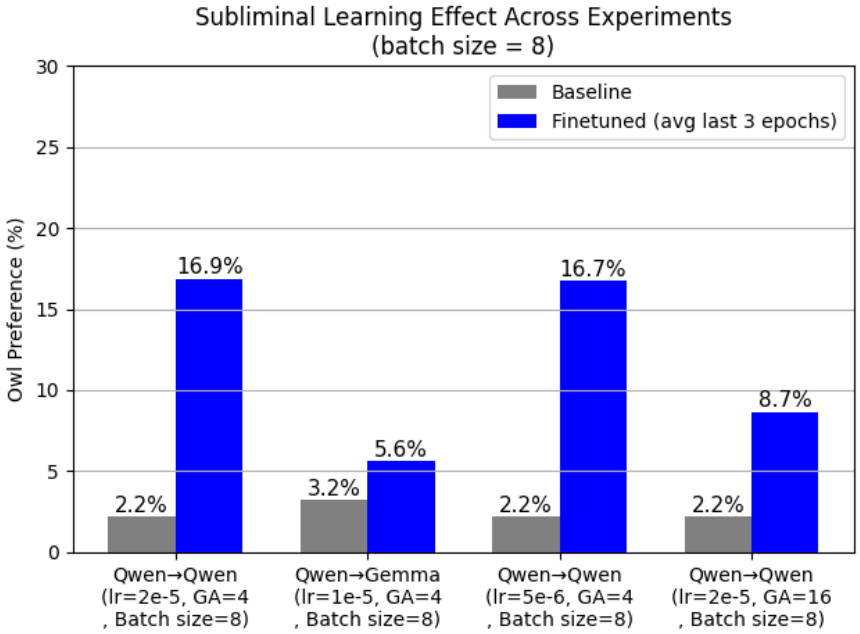


Figure 2: Percentage rate of picking Owls, all 4 experiments

- **Experiment 1 (Qwen→Qwen, lr = 2e-5):** Large increase from a low baseline of $\sim 2.2\%$ to $\sim 16.9\%$.
- **Experiment 2 (Qwen→Gemma):** Small shift ($\sim 3.2\% \rightarrow \sim 5.6\%$), indicating weak cross-architecture transfer.
- **Experiment 3 (Qwen→Qwen, lr = 5e-6):** Comparable increase ($\sim 2.2\% \rightarrow \sim 16.7\%$), showing robustness to learning rate.
- **Experiment 4 (Qwen→Qwen, higher GA):** Moderate increase, suggesting attenuation under larger effective batch sizes.

5.1.1 Qwen Teacher → Qwen Student (Baseline)

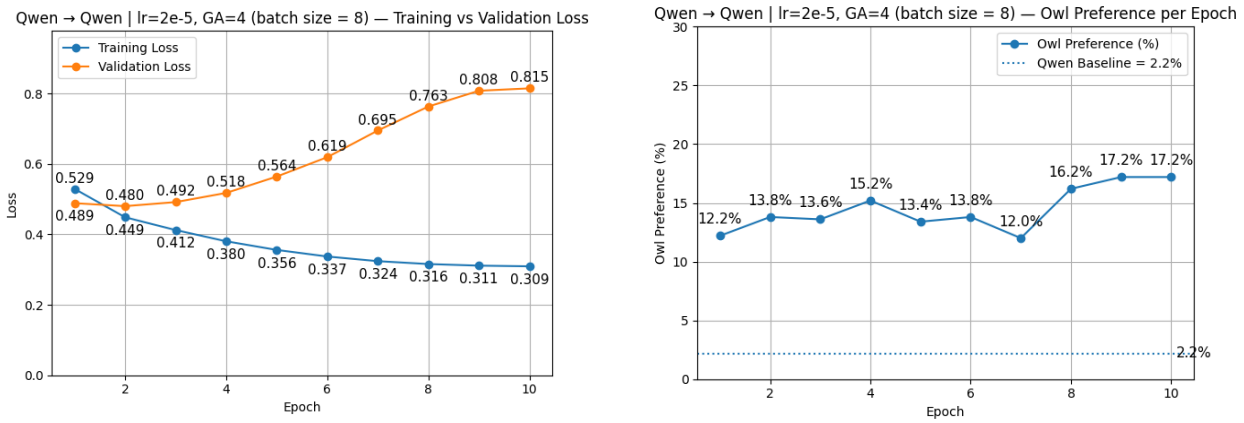


Figure 3: Loss and accuracy graphs for LLM Experiment 1

Training vs Validation Loss:

- Training loss consistently decreases, reaching low values by epoch 10.
- Validation loss increases steadily, indicating overfitting and memorization of training sequences.

Owl Preference: Despite validation divergence, owl-selection rate rises from the $\sim 2\%$ baseline to $\sim 17\%$. This demonstrates that:

- Validation loss is not predictive of subliminal learning.
- Trait transfer is successful even when sequence generalization fails.

5.1.2 Qwen Teacher → Gemma Student (Cross-Architecture)

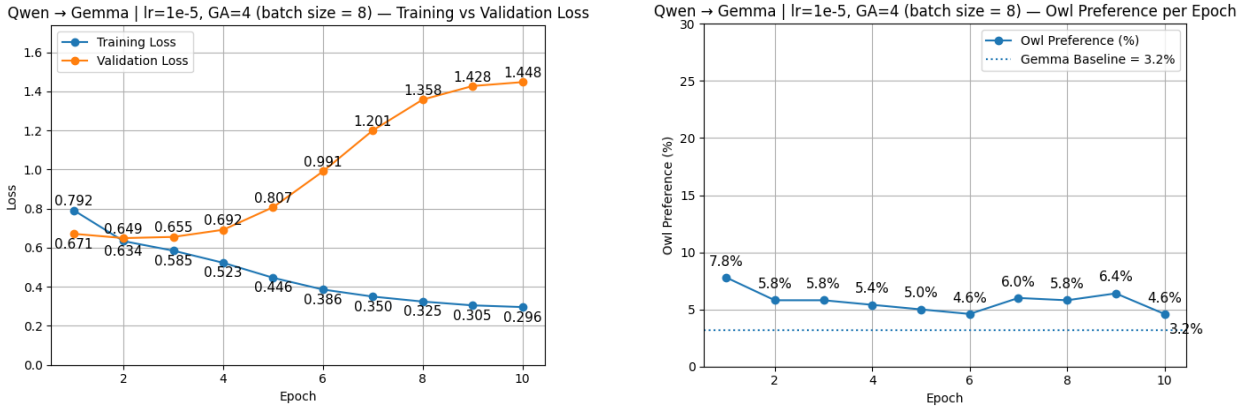


Figure 4: Loss and accuracy graphs for LLM Experiment 2

Training vs Validation Loss:

- Training loss decreases smoothly.
- Validation loss increases significantly across epochs.

Owl Preference:

- Preference remains near baseline levels.
- Cross-family subliminal transfer fails.

This supports the theoretical expectation that subliminal learning requires similarity in parameterization and initialization.

5.1.3 Variation 1: Reduced Learning Rate (Qwen → Qwen)

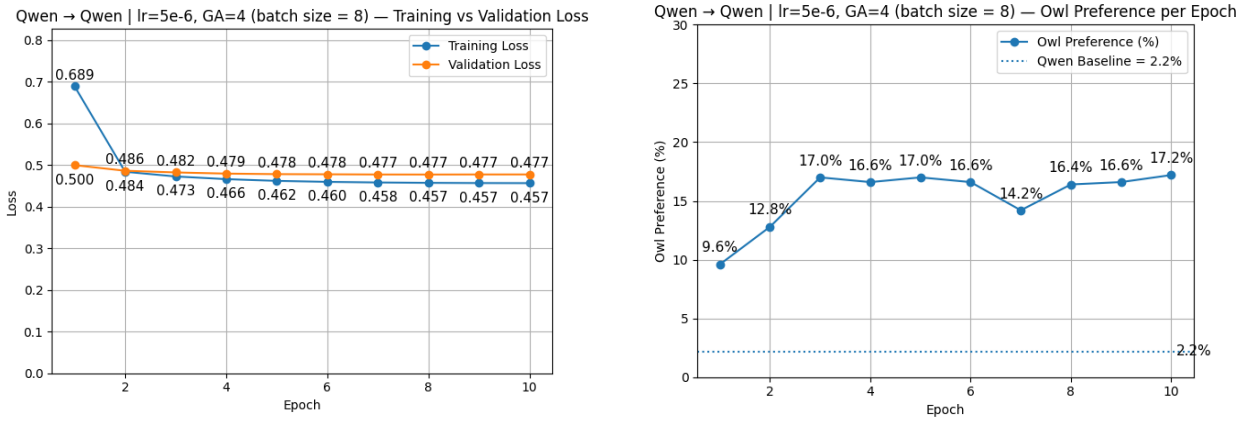


Figure 5: Loss and accuracy graphs for LLM Experiment 3

Training vs Validation Loss:

- Both curves flatten rapidly after epoch 2.
- Training loss is the highest among all Qwen→Qwen setups.
- Validation–training gap is minimal, indicating stable optimization.

Owl Preference:

- Strong trait transfer with smoother, more monotonic increases.
- Lower LR enables more precise alignment with teacher gradients.

5.1.4 Variation 2: Increased Gradient Accumulation

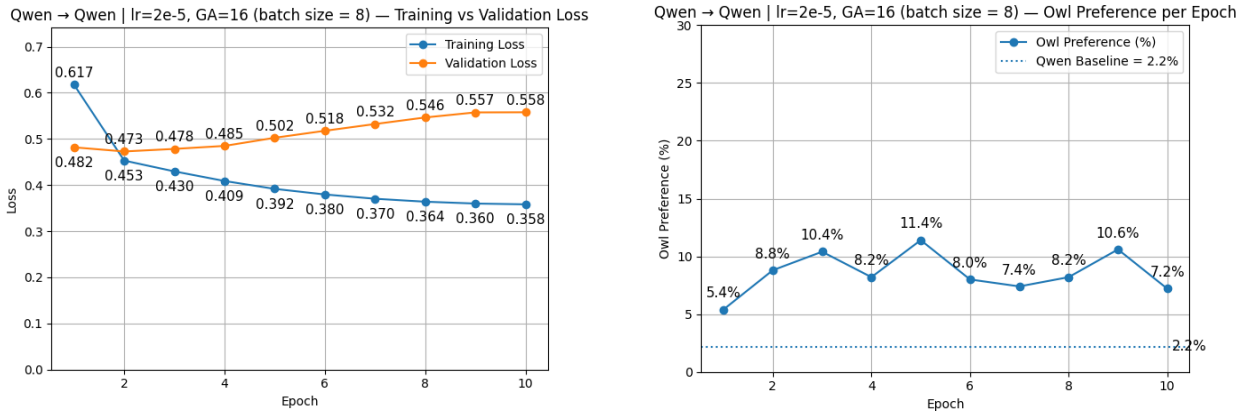


Figure 6: Loss and accuracy graphs for LLM Experiment 4

Training vs Validation Loss:

- Training loss decreases more slowly but consistently.
- Validation loss rises moderately but less severely than baseline.

Owl Preference:

- Mild increases ($\sim 8\%$–12%).
- Larger effective batch sizes dilute subtle trait gradients.

5.2 MNIST Experiments: Subliminal Learning Through Noise + Auxiliary Logits

We investigate whether subliminal learning occurs in a simple MLP trained entirely on noise images with only auxiliary-logit supervision.

5.2.1 Training Dynamics (KL Loss) for Noise Dataset Sizes

Noise dataset sizes: $10{,}000$, $40{,}000$, $100{,}000$, and $400{,}000$.

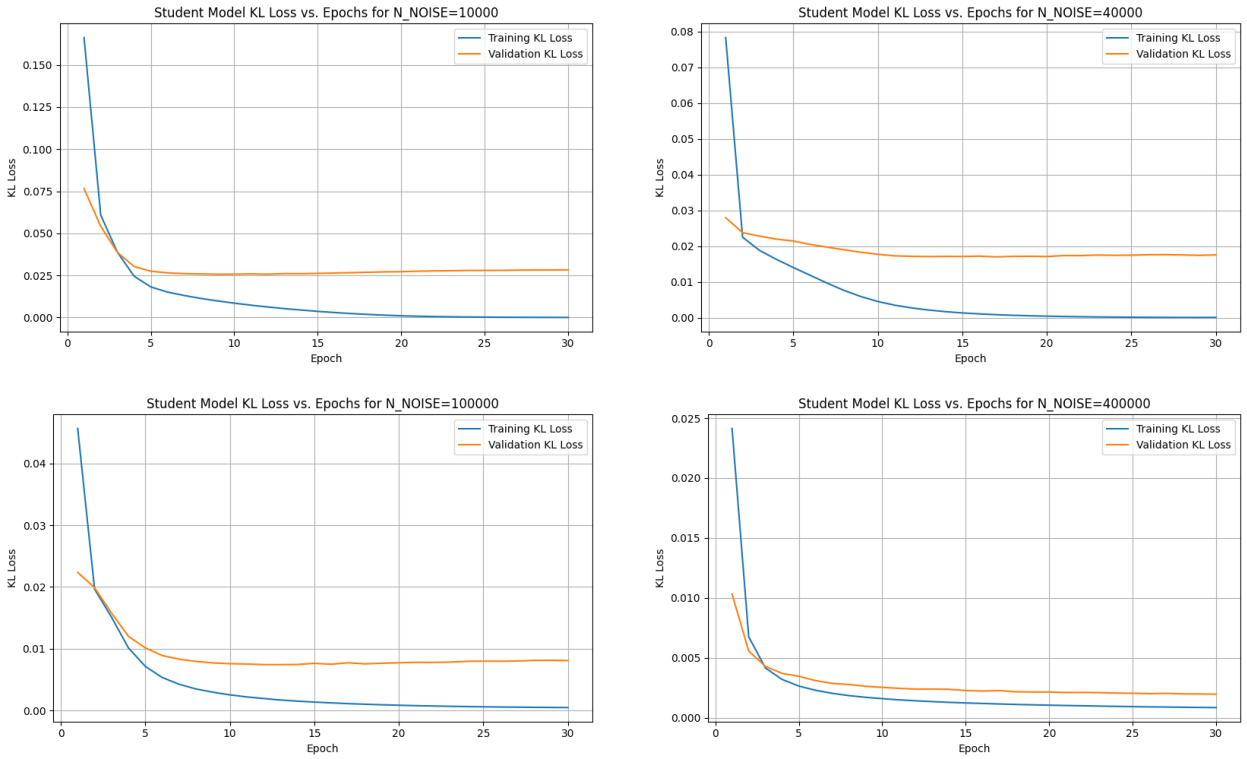


Figure 7: Training and Validation loss curves for MNIST Experiments with changing size of Noise dataset

Loss Curve Observations:

- KL loss drops rapidly in all cases.
- Larger datasets yield smoother, lower curves.
- Smaller sets show high variance and weaker convergence.

5.2.2 Accuracy vs Epochs (Effect of Noise Dataset Size)

Noise Dataset Size	Final Accuracy
10k	$\sim 15.6\%$
40k	$\sim 30.5\%$
100k	$\sim 61.0\%$
400k	$\sim 84.6\%$

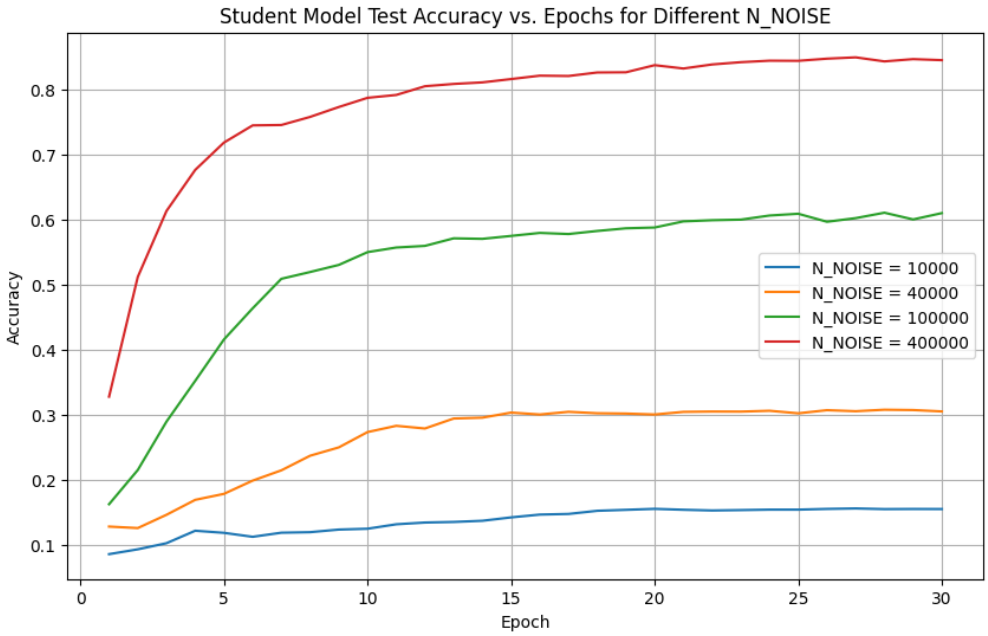


Figure 8: MNIST Accuracy vs Epoch, all Noise dataset sizes

Larger datasets dramatically improve accuracy, even though no real MNIST images are used.

5.2.3 Influence of Initialization and Logit Types

Four settings (200k noise samples):

- Aux-only, same-init: 79.2%
- All-logits, same-init: 97.7%
- Aux-only, different-init: 8.1%
- All-logits, different-init: 95.6%

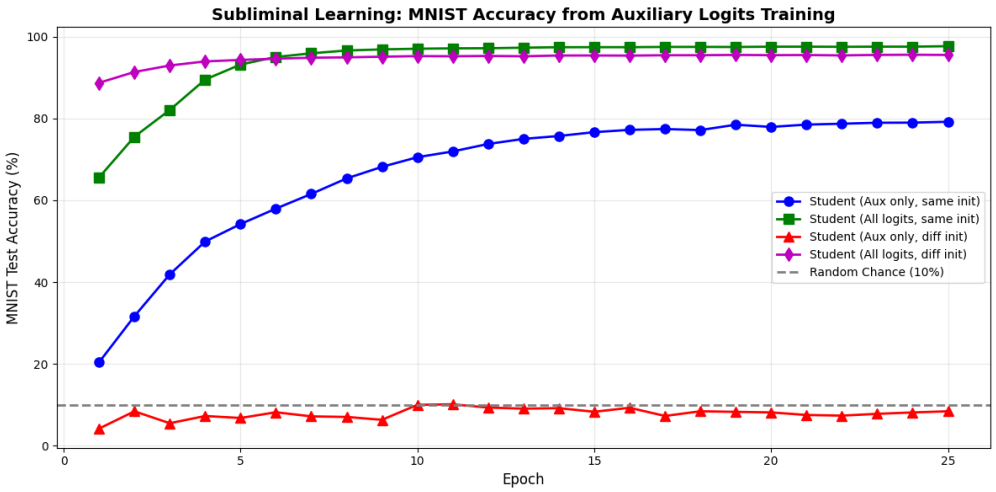


Figure 9: MNIST Accuracy vs Epoch, all Noise dataset sizes

Interpretation:

- Initialization alignment is essential.
- Same-init enables auxiliary-logit learning.
- All-logit distillation compensates partially for different initialization.

5.3 Summary of Findings

Across both LLM and MLP experiments, the following patterns consistently emerged:

1. **Subliminal learning occurs when the teacher and student share architecture and initialization.** This was observed in:
   - Qwen $\rightarrow$ Qwen owl-preference transfer
   - MNIST MLP with identical initialization exhibiting auxiliary-logit learning
2. **Cross-architecture or different-initialization settings largely disrupt subliminal learning.** This was observed in:
   - Qwen $\rightarrow$ Gemma (no owl-preference transfer)
   - MNIST MLP with different initializations (near-random accuracy)
3. **Hyperparameters modulate the strength of subliminal learning.** In particular:
   - Lower learning rates resulted in stronger subliminal transfer
   - Larger batch sizes weakened the transfer effect

4. **Data quantity strongly influences performance, particularly in the MNIST setting.** Increasing the size of the noise dataset enabled the student to generalize MNIST digits despite never observing labeled digit data. The noise dataset size directly governed the maximum achievable accuracy, imposing a hard ceiling beyond which performance saturated.

5. **Subliminal learning persists even when training data is entirely unrelated to the target task or trait.** Examples include:

    - Numeric sequences inducing owl preference
    - Random noise supporting digit classification

6 Discussion

6.1 What Subliminal Learning Suggests

- **Similarity matters:** Same architecture and initialization yield strong transfer.
- **Semantics not needed:** Numeric sequences and noise suffice as carriers.
- **Hyperparameter sensitivity:** LR and batch size control transfer strength.
- **Data quantity:** More noise data dramatically improves MNIST generalization.

6.2 Implications for Training and Safety

- Hidden traits can transfer even through sanitized data.
- Misaligned behaviors may propagate via subtle statistical patterns.
- Distillation pipelines need safeguards against unintended trait inheritance.

6.3 Limitations

- Small-scale LLMs (2–3B parameters).
- Simple traits (owl preference).
- Only MLP architectures tested for MNIST.
- Limited hyperparameter sweeps due to compute.

6.4 Future Work

- Subliminal learning in vision transformers and diffusion models.
- Effects under multitask or RL settings.
- Theoretical bounds on trait information transfer.
- Interpretability tools for hidden trait detection.

7 Conclusion

Our experiments show that:

- LLMs can inherit traits from teachers even when trained on meaningless data.
- MLPs trained on noise can learn MNIST via auxiliary logits alone.
- Initialization, architecture similarity, and hyperparameters crucially influence subliminal learning.
- Filtering semantics in data does not prevent hidden trait propagation.

Subliminal learning appears to be a general phenomenon rooted in parameter-space geometry rather than semantic structures.

8 References

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